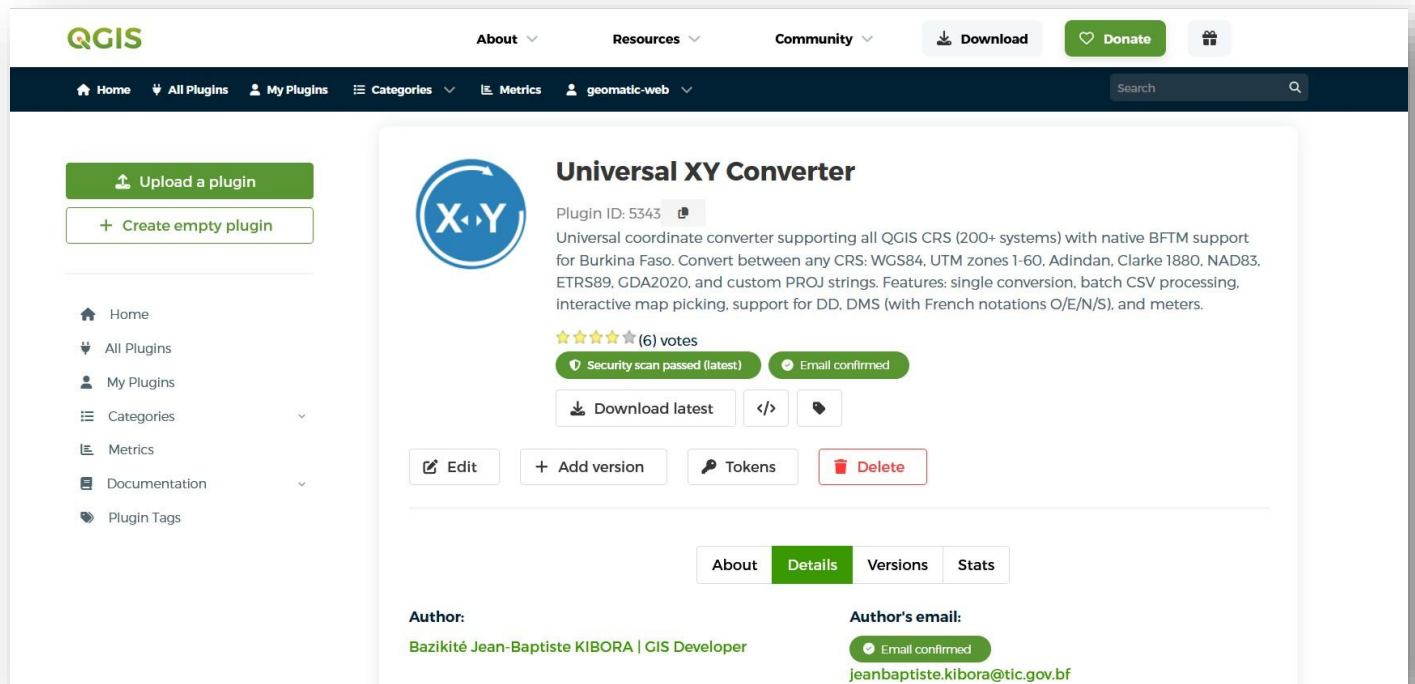


QGIS plugin - user guide

User Guide for the Universal XY Converter QGIS Plugin: Converting Coordinates Between Projection Systems, with Native BFTM Support



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Demo video: <https://youtu.be/PIOYuNvz2ZM>

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1. Introduction

Converting coordinates between reference systems is a common but often tedious GIS task, particularly for local systems that are poorly supported by generic tools. **Universal XY**

Converter simplifies this task directly within QGIS:

- Conversion between more than 200 coordinate systems
- Batch processing of CSV files (mass import/export)
- Interactive point selection on the map (click to retrieve coordinates)
- Native support for BFTM (Burkina Faso Transverse Mercator), a local projection system rarely available in standard CRS databases
- Custom CRS support via PROJ strings
- Multi-format export: CSV, GeoJSON, KML, Shapefile
- Conversion history for quick access

2. Requirements

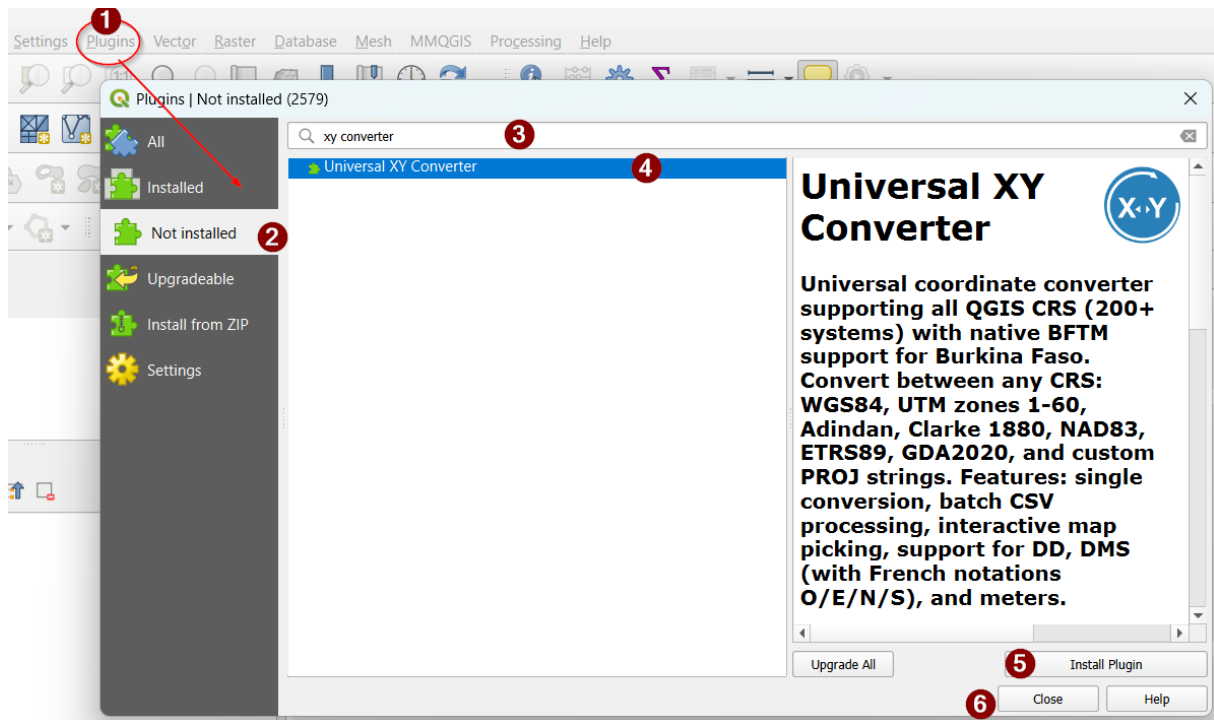
- **QGIS 3.0.0 or to QGIS 4.x**
- Basic familiarity with the coordinate systems used in your project (helpful but not required)

3. Installation

You have three methods to install the plugin:

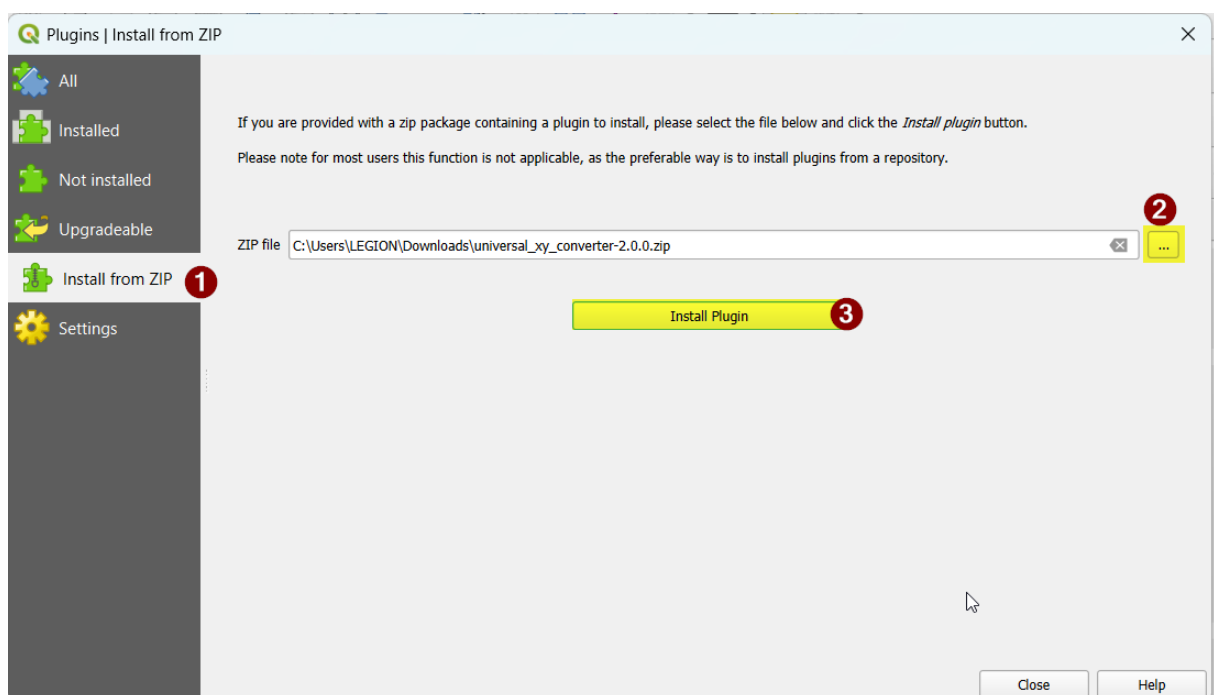
3.1. Method 1: From the official QGIS repository (recommended)

1. Open **QGIS**
2. Go to **Plugins → Manage and Install Plugins...**
3. Search for "**Universal XY Converter**"
4. Click **Install**



3.2. Method 2: Manual installation from a ZIP file

1. Download the ZIP file from:
https://plugins.qgis.org/plugins/universal_xy_converter/version/2.0.0/download/
2. In QGIS, go to **Plugins** → **Manage and Install Plugins...**
3. Click **Install from ZIP**
4. Select the ZIP file from your download folder
5. Confirm the installation



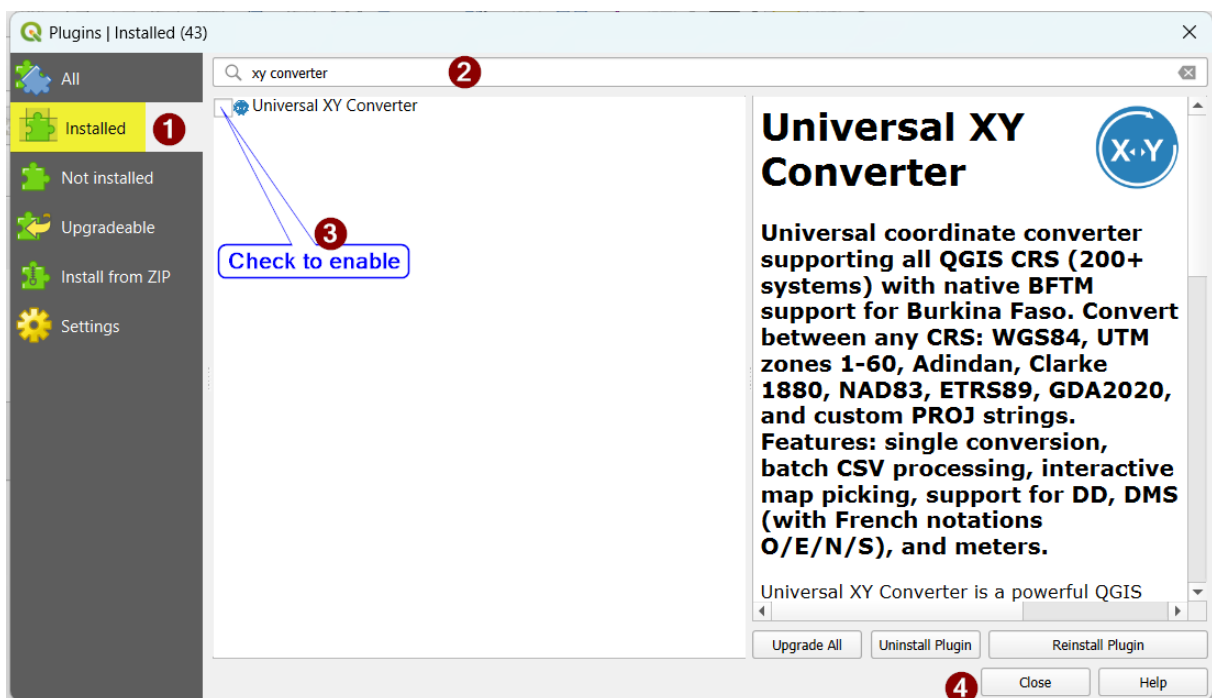
3.3. Method 3: From source code (advanced users)

Bash/cmd

```
# Clone the repository
git clone https://github.com/geomatic-web/Universal_xy_converter.git

# Copy the folder to the QGIS plugins directory
# (The path may vary depending on your operating system)
cp -r Universal_xy_converter
~/.local/share/QGIS/QGIS3/profiles/default/python/plugins/
```

Then enable the plugin in QGIS via **Plugins → Manage and Install Plugins → Installed**.



4. Interface Overview

The plugin interface is divided into **four main tabs**:

4.1. Tab 1: Simple Conversion

This tab allows you to convert a single point coordinate from one system to another.

- **Source CRS selection:** Choose the source coordinate system
- **Target CRS selection:** Choose the target coordinate system
- **Coordinate input:** Enter X/Y coordinates in your chosen format
- **Convert button:** Execute the conversion
- **Results display:** View converted coordinates with copy option

4.2. Tab 2: Batch Conversion

This tab allows you to convert multiple coordinates from a CSV file.

- **File selection:** Choose your CSV file
- **Column mapping:** Select X and Y columns
- **CRS configuration:** Set source and target systems
- **Processing options:** Keep original columns, skip invalid rows
- **Log display:** View processing progress and results

4.3. Tab 3: Interactive Selection

This tab allows you to pick coordinates directly from the QGIS map.

- **CRS selection:** Choose the target coordinate system
- **Format selection:** Choose output format (DD, DMS, or Meters)
- **Activate/Deactivate buttons:** Start and stop map selection mode
- **Results display:** View source coordinates and converted results

4.4. Tab 4: Help

This tab displays documentation and reference information.

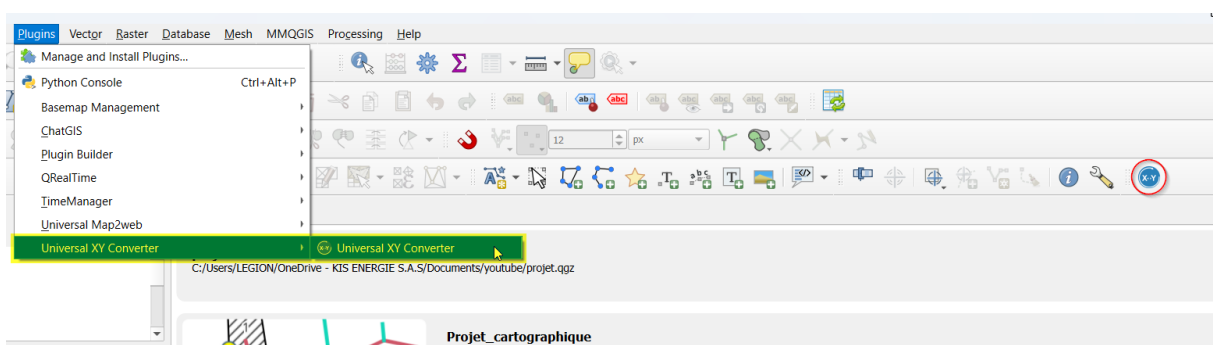
- **Coordinate format examples:** DD, DMS, and Meters formats
- **CSV format guide:** How to structure your input files
- **CRS reference:** Information about BFTM, ITRF2008, Clarke 1880, and Adindan
- **Support contacts:** Email and phone for assistance

5. Converting Coordinates

This function is ideal for converting a single point.

5.1. Manual Mode

Open the plugin: In the Plugins menu, point to **Universal XY Converter** and click the tool, or use its icon in the toolbar.



1. **Select source CRS:** Choose the reference system of the coordinates you want to convert from the dropdown list (among the 200+ available systems, including **BFTM**). You can search by name or by recognizing the system description.
2. **Select target CRS:** Choose the reference system you want to convert to (e.g., BFTM if working in Burkina Faso).

3. **Choose source format:** Select the format of your input coordinates:
 - **Decimal Degrees (DD):** e.g., 12.5042
 - **Degrees/Minutes/Seconds (DMS):** e.g., 12°30'15" N
 - **Meters:** e.g., 679246.00
4. **Enter coordinates:** Enter the X (or Longitude/Easting) and Y (or Latitude/Northing) coordinates in the provided fields.
5. **Choose target format:** Select the desired output format.
6. **Convert:** Click the **Convert** button. The new coordinates are displayed immediately.
7. **Copy results:** Click the **Copy** button to copy the results to your clipboard.

The screenshot shows the 'Universal XY Converter' application window. It has a title bar with a green icon and a close button. Below the title bar are four tabs: 'Simple Conversion' (selected), 'Batch Processing', 'Interactive Selection', and 'Help'. The main interface is divided into three sections: 'Source System', 'Target System', and 'Coordinates to Convert'. In the 'Source System' section, 'Source CRS' is set to 'WGS 84 (degrés)' (marked with a red circle 1) and 'Source Format' is set to 'Degrees/Minutes/Seconds (DMS)' (marked with a red circle 2). In the 'Target System' section, 'Target CRS' is set to 'WGS 84 / UTM zone 30N' (marked with a red circle 3) and 'Target Format' is set to 'Meters (m)' (marked with a red circle 4). In the 'Coordinates to Convert' section, the 'X / Longitude / East' field contains '1°30'12"O' (marked with a red circle 5) and the 'Y / Latitude / North' field contains '12°150'12"N' (marked with a red circle 6). Below these fields is a large green 'Convert →' button (marked with a red circle 7). A 'Swap source/target' button is located below the 'Convert' button. At the bottom, the 'Conversion Result' section shows 'X / East: 661284.370', 'Y / North: 1603921.276', and a status message '✓ Conversion successful'. A 'Copy results' button is at the bottom of this section. A red rectangle highlights the 'Conversion Result' section.

Example Conversions

- **Example 1: WGS84 to BFTM**

- Source: WGS84 (12.5042, -1.5)
- Target: BFTM
- Result: X: 679246.00, Y: 1380000.00

- **Example 2: DMS to Decimal Degrees**

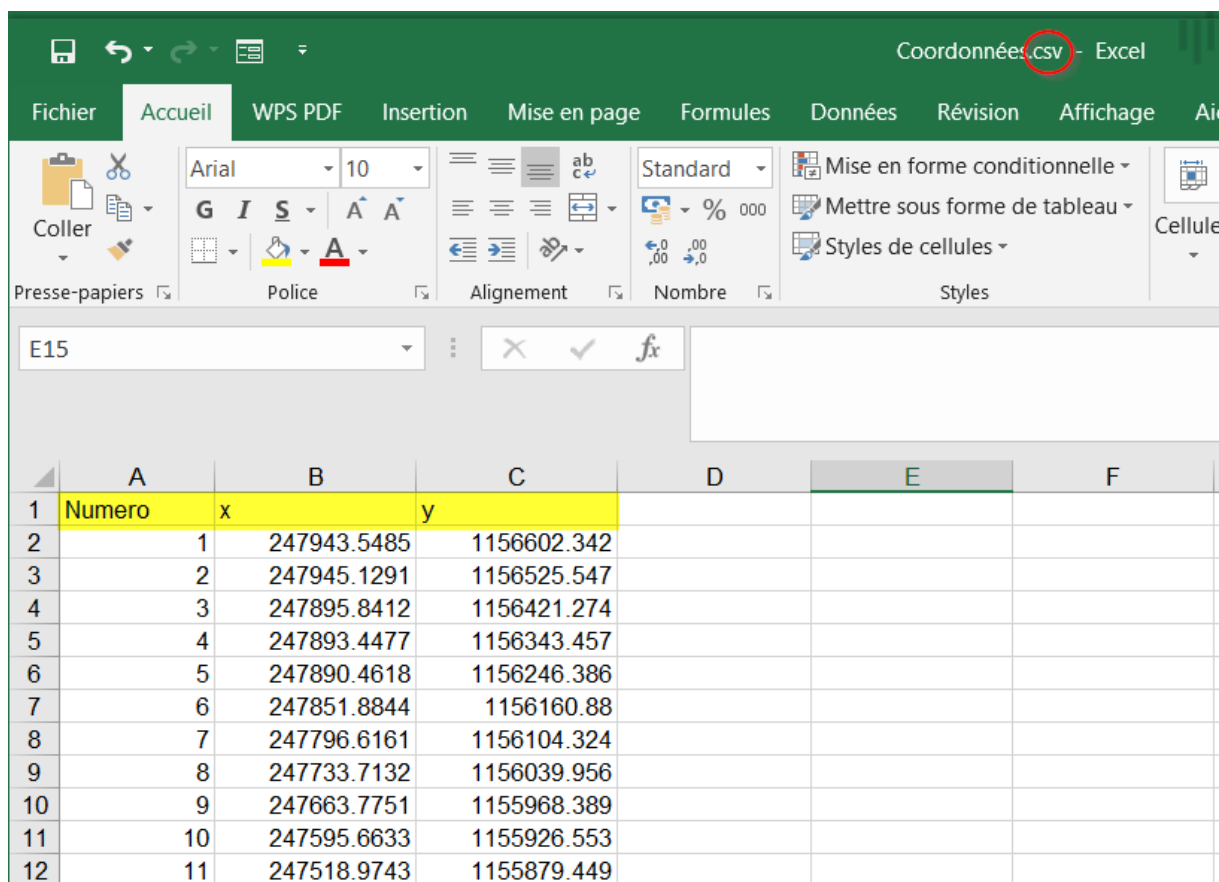
- Source: 12°30'15" N, 1°30'0" W
- Target: WGS84 (Decimal Degrees)
- Result: 12.5042, -1.5

5.2. Batch Processing (CSV Files)

This function is particularly useful for processing large volumes of data.

1. **Prepare your CSV file:** Make sure it contains at least two columns:

- One for X coordinates (longitude/Easting)
- One for Y coordinates (latitude/Northing)
- The first row must contain column headers



	A	B	C	D	E	F
1	Numero	x	y			
2	1	247943.5485	1156602.342			
3	2	247945.1291	1156525.547			
4	3	247895.8412	1156421.274			
5	4	247893.4477	1156343.457			
6	5	247890.4618	1156246.386			
7	6	247851.8844	1156160.88			
8	7	247796.6161	1156104.324			
9	8	247733.7132	1156039.956			
10	9	247663.7751	1155968.389			
11	10	247595.6633	1155926.553			
12	11	247518.9743	1155879.449			

2. **Select the 'Batch Conversion' tab:** Click on the Batch Conversion tab.
3. **Load the file:** Click "Browse" and select your CSV file. The plugin will automatically detect column headers.
4. **Configure columns:** Specify:
 - The column containing X coordinates
 - The column containing Y coordinates
 - The separator used in your CSV (comma, semicolon, tab, or pipe)
5. **Set source CRS:** Select the Coordinate Reference System (CRS) of your data.
6. **Set source format:** Choose the format of your input coordinates (DD, DMS, or Meters).
7. **Set target CRS:** Select the target CRS for your new coordinates.
8. **Set output options:** Choose how the new coordinates will be added:
 - **Keep original columns:** Add new columns at the end of the file
 - **Skip invalid rows:** Ignore rows with invalid coordinates
9. **Run conversion:** Click **Process**. The plugin will:
 - Process each row
 - Show progress in the progress bar
 - Display detailed logs in the log window
 - Create a new CSV file with the converted coordinates

Universal XY Converter

Simple Conversion | **Batch Processing** | Interactive Selection | Help

1. Source File
 C:/Users/LEGION/Downloads/Coordonnées.csv **2** Browse

2. Column Configuration
 X / Longitude column: **x** **3**
 Y / Latitude column: **y** **4**
 CSV Separator: **;** **5**

3. Coordinate Systems
 Source system: **WGS 84 / UTM zone 30N** **6**
 Source format: **Meters (m)**
 Target system: **BFTM (Burkina Faso) - OFFICIEL** **7**

4. Options
☒ Keep original columns
☒ Skip invalid rows

5. Start Conversion **8**

Log
 Format: Meters (m)
 BATCH CONVERSION
 File: Coordonnées.csv
 Format: Meters (m)
 ===== **9**
 CONVERSION REPORT
 =====
 Converted rows: 240
 File: Coordonnées_converti.csv
 =====
 Completed: C:/Users/LEGION/Downloads/Coordonnées_converti.csv

5.2.1. Output Options

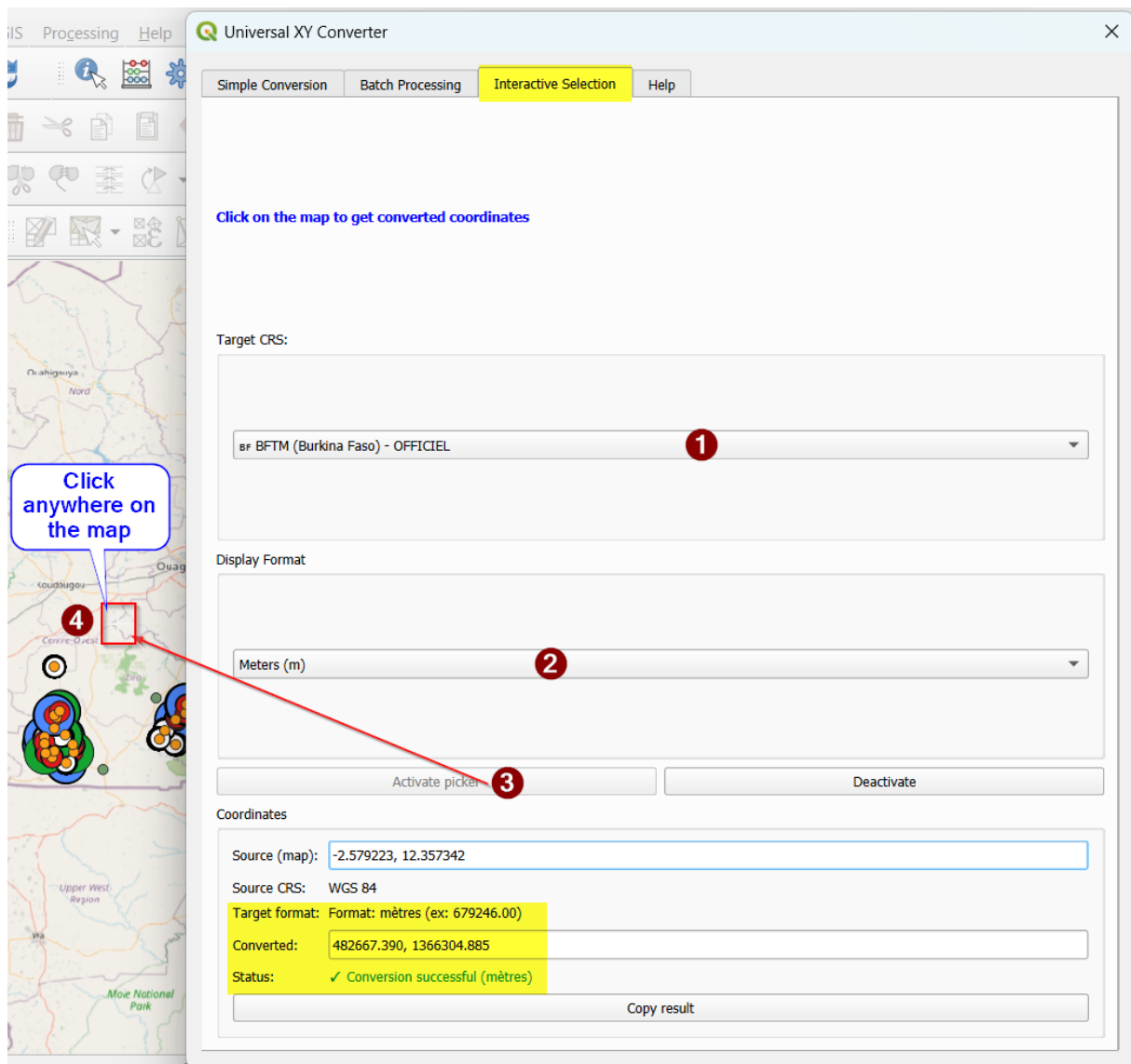
Option	Description
Keep original columns	Adds converted coordinates as new columns (e.g., <code>x_converted</code> , <code>y_converted</code>)
Skip invalid rows	Skips rows with invalid coordinates and logs the errors

	A	B	C	D	E	F
	Numero	x	y	X_CONVERTI	Y_CONVERTI	
1						
2	1	247943.549	1156602.34	183560.1854	1158193.245	
3	2	247945.129	1156525.55	183561.4018	1158116.339	
4	3	247895.841	1156421.27	183511.5501	1158012.16	
5	4	247893.448	1156343.46	183508.7822	1157934.25	
5	5	247890.462	1156246.39	183505.3293	1157837.062	
7	6	247851.884	1156160.88	183466.2921	1157751.624	
8	7	247796.616	1156104.32	183410.6795	1157695.256	
9	8	247733.713	1156039.96	183347.3848	1157631.101	
0	9	247663.775	1155968.39	183277.0111	1157559.771	
1	10	247595.663	1155926.55	183208.6079	1157518.203	
2	11	247518.974	1155879.45	183131.5908	1157471.401	

5.3. Interactive Selection on the Map

This function allows you to retrieve the coordinates of a point by clicking directly on the QGIS map.

1. **Open the plugin:** Launch *Universal XY Converter*.
2. **Select the 'Pick from Map' tab:** Click on the Interactive Selection tab.
3. **Choose output CRS:** Select the reference system in which you want to get the coordinates of the captured point (e.g., WGS84 or BFTM).
4. **Choose output format:** Select how you want the coordinates displayed (DD, DMS, or Meters).
5. **Activate map selection:** Click the **Activate** button. The cursor will change to a crosshair.
6. **Click on the map:** Click on any point in the QGIS map window. The coordinates of the clicked point are automatically:
 - o Retrieved in the project's current CRS
 - o Converted to your chosen target CRS
 - o Displayed in your chosen format
7. **Copy results:** Click the **Copy** button to copy the converted coordinates to your clipboard.
8. **Deactivate:** Click the **Deactivate** button to exit selection mode.



6. Using a Custom CRS (PROJ String)

If the system you're looking for isn't in the list, or if you work with a highly specific reference system:

6.1. Step-by-Step Instructions

1. **Select the Custom CRS option:** In the CRS dropdown, select "--- Personnalisé (PROJ) ---" (Custom).
2. **Enter the PROJ string:** Type or paste the PROJ string defining your projection system.
3. **Use the custom CRS:** The custom CRS will be available for conversion, just like a standard system.

Universal XY Converter

Simple Conversion | Batch Processing | Interactive Selection | Help

Source System

Source CRS: --- Personnalisé (PROJ) --- (1)

Source Format: Degrees/Minutes/Seconds (DMS)

Source PROJ string: ex: +proj=longlat +ellps=clrk80 +towgs84=-118,-14,218 (2)

Target System

Target CRS: WGS 84 / UTM zone 30N (3)

Target Format: Meters (m) (4)

Coordinates to Convert

X / Longitude / East: 1°30'12"O (5)

Y / Latitude / North: 12°150'12"N (6)

Convert → (7)

↔ Swap source/target

Conversion Result

X / East: 661284.370

Y / North: 1603921.276

Status: ✓ Conversion successful

Copy results

6.2. PROJ String Examples

6.2.1. BFTM (Burkina Faso Transverse Mercator)

text

```
+proj=tmerc +lat_0=0 +lon_0=-1.5 +k=0.9996 +x_0=600000 +y_0=0 +ellps=GRS80
+units=m +no_defs
```

6.2.2. Clarke 1880 (West Africa)

text

```
+proj=longlat +ellps=clrk80 +towgs84=-118,-14,218 +no_defs
```

6.2.3. ITRF2008

text

```
+proj=longlat +ellps=GRS80 +towgs84=0,0,0,0,0,0,0 +no_defs
```

7. The BFTM System (Burkina Faso Transverse Mercator)

BFTM is built into the plugin natively, with no additional configuration required. It appears directly in the list of available coordinate systems, both as source and target. This allows users working with data in Burkina Faso to convert directly to/from this local reference system without manually defining a PROJ string.

7.1.1. Why BFTM Matters

- **Official projection system** for Burkina Faso
- **Accurate local measurements** for the region
- **Rarely available** in generic tools
- **Native support eliminates** manual configuration errors

7.1.2. BFTM Technical Specifications

Parameter	Value
Projection	Transverse Mercator
Ellipsoid	GRS80
DATUM	ITRF 2008
Central meridian	1.5° West
False easting	600,000 m
False northing	0 m
Scale factor	0.9996
Units	Meters

8. Coordinate Formats Supported

The plugin supports three main coordinate formats:

8.1. Decimal Degrees (DD)

Description: Coordinates expressed as decimal numbers.

Examples:

- Latitude: 12.5042 (positive = North)
- Longitude: -1.5 (negative = West)

Format: 12.5042, -1.5

8.2. Degrees/Minutes/Seconds (DMS)

Description: Coordinates expressed in degrees, minutes, and seconds with directional indicators.

Examples:

- Latitude: 12°30'15" N
- Longitude: 1°30'0" W

Format: 12°30'15" N, 1°30'0" W

Direction indicators:

- **N:** North (latitude positive)
- **S:** South (latitude negative)
- **E:** East (longitude positive)
- **W/O:** West (longitude negative)

8.3. Meters (Projected)

Description: Coordinates expressed in meters (projected systems like UTM or BFTM).

Examples:

- X (Easting): 679246.00
- Y (Northing): 1380000.00

Format: 679246.00, 1380000.00

9. Troubleshooting / FAQ

9.1. Q: My coordinate system doesn't appear in the list.

A: Use the **Custom CRS** option and enter the corresponding PROJ string. The plugin supports over 200 systems, but you may need to add a custom one for very specific projections.

9.2. Q: Batch conversion fails on my CSV file.

A: Check that:

- The coordinate columns are correctly named and mapped
- The format is correct (e.g., no comma as decimal separator if the CSV expects a period)
- The CSV uses UTF-8 encoding
- There are no empty rows or invalid characters

9.3. Q: BFTM gives inconsistent results.

A: Make sure the source system of your data actually matches the expected reference system before conversion. A mismatched source CRS is the most common cause of positional errors. Verify your input coordinates are in the correct system.

9.4. Q: Can I retrieve an older conversion?

A: Yes, via the built-in conversion history. Click on any previous entry to reload its settings.

9.5. Q: Why is my exported file not opening correctly?

A: Verify that:

- You selected the correct output format
- Your software supports the chosen format
- For Shapefile export, all required files (.shp, .shx, .dbf) are preserved

9.6. Q: Can I use this plugin offline?

A: Yes, the plugin works entirely offline once installed. No internet connection is required for conversions.

9.7. Q: What coordinate systems are supported?

A: The plugin supports over 200 QGIS projection systems, including:

- All major global systems (WGS84, UTM zones, etc.)
- Local systems like BFTM
- Historical systems like Clarke 1880 and Adindan
- Custom systems added via PROJ strings

9.8. Q: What is the difference between Decimal Degrees and DMS?

A: Decimal Degrees (DD) express coordinates as decimal numbers (e.g., 12.5042). DMS express coordinates in degrees, minutes, and seconds (e.g., 12°30'15" N). Both represent the same location; the plugin can convert between formats.

9.9. Q: Can I convert between projected and geographic systems?

A: Yes, the plugin can convert between any supported systems, including between projected (e.g., BFTM, UTM) and geographic (e.g., WGS84) systems.

9.10. Q: What does "Skip invalid rows" do in batch conversion?

A: When enabled, the plugin will skip rows that contain invalid coordinates (e.g., non-numeric values, empty fields) and continue processing the rest. Invalid rows are logged in the log window.

9.11. Q: How accurate are the conversions?

A: The conversions use QGIS's coordinate transformation engine, which provides high accuracy based on the parameters of each CRS. Accuracy depends on the transformation parameters available for each CRS pair.

10. Technical Reference

- **Clarke 1880 Ellipsoid**
 - Historical ellipsoid **used in Africa**
 - Semi-major axis: **6,378,249.145 m**
 - Flattening: **1/293.465**
 - To WGS84 (West Africa): **-118, -14, 218**
 - To WGS84 (Cameroon): **-166, -15, 204**
- **Adindan Datum**
 - Datum used in Burkina Faso and West Africa
 - Ellipsoid: Clarke 1880
 - To WGS84 (West Africa): -118, -14, 218
 - To WGS84 (Cameroon): -166, -15, 204
- **ITRF2008**
 - International Terrestrial Reference Frame 2008
 - Ellipsoid: GRS80 (a=6378137.0, f=1/298.257222101)
 - Prime Meridian: Greenwich
 - Angular Unit: Degree (0.0174532925199433)
 - EPSG: 8999

11. Appendix

11.1. Quick Reference Card

Task	Steps
Single point conversion	Select source CRS → Enter X/Y → Select target CRS → Click Convert
Map click conversion	Open Pick from Map tab → Click Activate → Click on map → View converted coordinates
Batch CSV conversion	Load CSV → Map columns → Set CRS → Click Process → Export

Add custom CRS	Select "Custom" in CRS dropdown → Enter PROJ string → Use as source/target
Reuse previous conversion	Open History → Click on desired entry → Parameters reloaded
Export results	Choose format (CSV/GeoJSON/KML/Shapefile) → Click Export
Copy coordinates	Click Copy button → Paste into any application
Swap CRS	Click Swap button → Source and target CRS exchanged

11.2. Common CRS Codes

System	Description	EPSG/ID
WGS84	World Geodetic System 1984	EPSG:4326
WGS84 Web Mercator	Web mapping standard	EPSG:3857
BFTM	Burkina Faso Transverse Mercator	Custom (built-in)
ITRF2008	International Terrestrial Reference Frame	EPSG:8999
UTM zones	Universal Transverse Mercator	EPSG:32601-32660 (North) / 32701-32760 (South)
Clarke 1880	Historical West Africa	Custom (PROJ)
Adindan	West Africa datum	Custom (PROJ)

11.3. PROJ String Reference

- **BFTM (Burkina Faso Transverse Mercator)**

text

```
+proj=tmerc +lat_0=0 +lon_0=-1.5 +k=0.9996 +x_0=600000 +y_0=0 +ellps=GRS80 +units=m +no_defs
```

- **Clarke 1880 (West Africa)**

text

```
+proj=longlat +ellps=clrk80 +towgs84=-118,-14,218 +no_defs
```

- **Clarke 1880 (Cameroon)**

text

```
+proj=longlat +ellps=clrk80 +towgs84=-166,-15,204 +no_defs
```

- **ITRF2008**

text

```
+proj=longlat +ellps=GRS80 +towgs84=0,0,0,0,0,0,0 +no_defs
```

- **Adindan (West Africa)**

text

```
+proj=longlat +ellps=clrk80 +towgs84=-118,-14,218 +no_defs
```

- **UTM Zone 30N (WGS84)**

text

```
+proj=utm +zone=30 +datum=WGS84 +units=m +no_defs
```